

CYBERSECURITY

What is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, software, devices, and data from unauthorized access, cyberattacks, damage, or theft. It helps individuals and organizations protect sensitive information and maintain the security of their digital systems.

Importance of Cybersecurity

Cybersecurity is important because people and organizations store a large amount of personal, financial, business, and confidential information digitally. Cyberattacks can lead to data theft, financial losses, privacy violations, system damage, and interruption of services.

CIA Triad

The CIA Triad is a fundamental concept in cybersecurity. It consists of:

1. **Confidentiality:** Confidentiality ensures that information is accessible only to authorized users. Examples: Passwords, encryption, access controls, and authentication.
2. **Integrity:** Integrity ensures that information remains accurate, complete, and protected from unauthorized modification. Examples: Hashing, digital signatures, file permissions, and version control.
3. **Availability:** Availability ensures that authorized users can access systems and information whenever they are required. Examples: Backups, redundant systems, disaster recovery, and protection against denial-of-service attacks.

10 COMMON CYBER THREATS

Phishing – Fraudulent emails, messages, or websites designed to trick users into revealing sensitive information such as passwords or banking details.

Malware – Malicious software designed to damage systems, steal information, or gain unauthorized access. Examples include viruses, worms, trojans, and spyware.

Ransomware – Malware that encrypts or blocks access to files or systems and demands payment from the victim.

Password Attacks – Attempts to obtain or crack passwords through methods such as brute-force attacks, password guessing, or credential theft.

Denial-of-Service (DoS/DDoS) Attacks – Attacks that overwhelm a system or network with traffic or requests, making services unavailable to legitimate users.

Social Engineering – Manipulating people into revealing confidential information or performing actions that compromise security.

Man-in-the-Middle (MitM) Attacks – An attacker secretly intercepts communication between two parties to steal or manipulate information.

SQL Injection – An attack in which malicious SQL commands are inserted into application inputs to access or manipulate database information.

Insider Threats – Security threats caused by people within an organization who misuse their authorized access, either intentionally or accidentally.

Zero-Day Attacks – Attacks that exploit previously unknown or unpatched vulnerabilities before an effective security fix is available.